

ECMT676 Econometrics II
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Tentative Solutions to Problem 5

1. (15.6)

- (a) What we would like to know is that, if we exogenously change the number of cigarettes that someone smokes per day, what effect would this have on the probability of missing work over a three-month period? In other words, we want to infer causality, not just find a correlation between missing work and cigarette smoking.
- (b) Since people choose whether and how much to smoke, we certainly cannot treat the data as coming from the experiment we have in mind in part (a). (That is, we cannot randomly assign people a daily cigarette consumption.) It is possible that smokers are less healthy to begin with, or have other attributes that cause them to miss work more often. Or, it could go the other way: cigarette consumption may be related to personality traits that make people harder workers. In any case, *cigs* might be correlated with the unobservables in the equation.
- (c) If we start with $P(y = 1|\mathbf{z}, \text{cigs}, q_1) = \Phi(\mathbf{z}_{i1}\delta_1 + \gamma_1 \text{cigs} + q_1)$, but ignore q_1 when it is correlated with *cigs*, we will not consistently estimate anything of interest, whether the model is linear or non-linear. Thus, we would not be estimating a causal effects. If q_1 is independent of *cigs*, the probit ignoring q_1 does estimate the average partial effect of another cigarette.
- (d) No. There are many people in the working population who do not smoke. Thus, the distribution (conditional or unconditional) of *cigs* piles up at zero. Also, since *cigs* takes on integer values, it cannot be normally distributed. But it is really the pile up at zero that is the most serious issue.
- (e) Use the Rivers-Vuong test. Obtain the residuals, $\hat{r}_2$, from the regressions *cigs* on $\mathbf{z}_1$, *cigs*, $\hat{r}_2$ and use a standard t test on $\hat{r}_2$. This does not rely on normality of r_2 (or *cigs*). It does, of course, rely on the probit model being correct for y under H_0 .
- (f) Assuming people will not immediately move out of their state of residence when the state implements no smoking laws in the workplace, and that state of residence is roughly independent of general health in the population, a dummy indicator for whether the person works in a state with a new law can be treated as exogenous and excluded from (15.98). (These situations are often called "natural experiments.") Further, *cigs* is likely to be correlated with the state law indicator because since people will not be able to smoke as much as they otherwise would. Thus, it seems to be a reasonable instrument for *cigs*.

2. (15.12) Follow the similar approach

$$P(y_{i1} = 1, y_{i2} = 0, y_{i3} = 0 | \mathbf{x}_i, c_i, n_i) = \frac{P(y_{i1} = 1, y_{i2} = 0, y_{i3} = 0 | \mathbf{x}_i, c_i)}{P(n_i | \mathbf{x}_i, c_i)}$$

and by conditional independence,

$$\begin{aligned} P(y_{i1} = 1, y_{i2} = 0, y_{i3} = 0 | \mathbf{x}_i, c_i, n_i) &= P(y_{i1} = 1 | \mathbf{x}_i, c_i, n_i) P(y_{i2} = 0 | \mathbf{x}_i, c_i, n_i) P(y_{i3} = 0 | \mathbf{x}_i, c_i, n_i) \\ &= \Lambda(\mathbf{x}'_{i1}\boldsymbol{\beta} + c_i) [1 - \Lambda(\mathbf{x}'_{i2}\boldsymbol{\beta} + c_i)] [1 - \Lambda(\mathbf{x}'_{i3}\boldsymbol{\beta} + c_i)] \end{aligned}$$

and

$$\begin{aligned} P(n_i = 1 | \mathbf{x}_i, c_i) &= P(y_{i1} = 1, y_{i2} = 0, y_{i3} = 0 | \mathbf{x}_i, c_i, n_i) \\ &\quad + [1 - \Lambda(\mathbf{x}'_{i1}\boldsymbol{\beta} + c_i)] \Lambda(\mathbf{x}'_{i2}\boldsymbol{\beta} + c_i) [1 - \Lambda(\mathbf{x}'_{i3}\boldsymbol{\beta} + c_i)] \\ &\quad + [1 - \Lambda(\mathbf{x}'_{i1}\boldsymbol{\beta} + c_i)] [1 - \Lambda(\mathbf{x}'_{i2}\boldsymbol{\beta} + c_i)] \Lambda(\mathbf{x}'_{i3}\boldsymbol{\beta} + c_i). \end{aligned}$$

Thus,

$$P(y_{i1} = 1, y_{i2} = 0, y_{i3} = 0 | \mathbf{x}_i, c_i, n_i) = \frac{e^{\mathbf{x}'_{i1}\beta}}{\sum_{k=1}^3 e^{\mathbf{x}'_{ik}\beta}}$$

3. (15.18)

- (a) $P(y_{it} = 1 | \mathbf{x}_i, a_i) = \Phi(\psi + \mathbf{x}_{it}\boldsymbol{\beta} + \bar{\mathbf{x}}_i\zeta + a_i)$. The fact that a_i given $\mathbf{x}_i$ is heteroskedastic has no bearing on the distribution conditional on $(\mathbf{x}_i, a_i)$. Only when we "integrate out" a_i does $D(a_i | \mathbf{x}_i)$ matter.
- (b) Let $f(y_{it} | \mathbf{x}_i, a_i) = [\Phi(\psi + \mathbf{x}_{it}\boldsymbol{\beta} + \bar{\mathbf{x}}_i\zeta + a_i)]^{y_{it}} [1 - \Phi(\psi + \mathbf{x}_{it}\boldsymbol{\beta} + \bar{\mathbf{x}}_i\zeta + a_i)]^{1-y_{it}}$. Then, by the product and integration rules,

$$f(y_i | \mathbf{x}_i, \delta) = f(y_{i1}, \dots, y_{iT} | \mathbf{x}; \theta) = \int_{-\infty}^{\infty} \left(\prod_{t=1}^T f(y_{it} | \mathbf{x}_i, a_i; \theta) \right) h(a_i | \mathbf{x}_i; \delta) da_i,$$

where $h(\cdot | \mathbf{x}_i, \delta)$ is $N[0, \sigma_a^2 e^{\bar{\mathbf{x}}_i\lambda}]$. Then the log-likelihood is

$$\sum_i \ln f(y_i | \mathbf{x}_i, \delta).$$

- (c) To show that $E[\Phi(\mathbf{x}^\circ\boldsymbol{\beta} + \psi + \bar{\mathbf{x}}_i\zeta + a_i) | \mathbf{x}_i] = \Phi([\mathbf{x}^\circ\boldsymbol{\beta} + \psi + \bar{\mathbf{x}}_i\zeta] / [1 + \sigma_a^2 e^{\bar{\mathbf{x}}_i\lambda}]^{1/2})$, we first

$$\begin{aligned} E[\Phi(\mathbf{x}^\circ\boldsymbol{\beta} + \psi + \bar{\mathbf{x}}_i\zeta + a_i) | \mathbf{x}_i] &= E\{E[\mathbb{I}\{\mathbf{x}^\circ\boldsymbol{\beta} + \psi + \bar{\mathbf{x}}_i\zeta + a_i + u_{it} > 0\} | \mathbf{x}_i]\} \\ &= E\left[\Phi\left(\frac{\mathbf{x}^\circ\boldsymbol{\beta} + \psi + \bar{\mathbf{x}}_i\zeta}{[1 + \sigma_a^2 e^{\bar{\mathbf{x}}_i\lambda}]^{1/2}}\right) | \mathbf{x}_i\right] = \Phi\left(\frac{\mathbf{x}^\circ\boldsymbol{\beta} + \psi + \bar{\mathbf{x}}_i\zeta}{[1 + \sigma_a^2 e^{\bar{\mathbf{x}}_i\lambda}]^{1/2}}\right). \end{aligned}$$

Therefore, a consistent estimator is obtained by using a sample average and plugging in the ML estimators and take derivatives and chagnes with respects to $\mathbf{x}^\circ$.

4. (17.18)

- (a) Once we assume $\mathbf{z}$ is exogenous in the structural equation — and $E(u_1 | \mathbf{z}) = 0$ ensures exogeneity — then we only need the rank condition. The assumption that $E(\mathbf{z}\mathbf{z}')$ is nonsingular is not usually restrictive. The important condition with a single endogenous explanatory variable is $L(y_2 | \mathbf{z}) \neq L(y_2 | \mathbf{z}_1)$, so there is at least one element of $\mathbf{z}$ not in $\mathbf{z}_1$ that explains variation in y_2 .
- (b) We can draw the optimal instrument variables results from Section 8.6. The condition $E(u_1 | \mathbf{z}) = 0$ ensures that any function of $\mathbf{z}$ is a valid instrumental variable candidate, and also implies that $E(u_1^2 | \mathbf{z}) = \text{var}(u_1 | \mathbf{z})$. Because $E(u_1^2 | \mathbf{z})$ is constant, from Theorem 8.5 the optimal IVs are $[E(y_2 | \mathbf{z}), \mathbf{z}_1]$. If we think $D(y_2 | \mathbf{z})$ follows a standard Tobit then obtain $E(y_2 | \mathbf{z})$ from the Tobit model. Recall that if $D(y_2 | \mathbf{z}) = \text{Tobit}(\mathbf{z}\delta_2, \tau_2^2)$ then $E(y_2 | \mathbf{z}) = \Phi(\mathbf{z}\delta_2 / \tau_2) \mathbf{z}\delta_2 + \tau_2 \phi(\mathbf{z}\delta_2 / \tau_2)$. Therefore, if we run Tobit in a first stage, we get $\hat{m}_{i2} \equiv \hat{E}(y_{i2} | \mathbf{z}_i) = \Phi(\mathbf{z}_i \hat{\delta}_2 / \hat{\tau}_2) \mathbf{z}_i \hat{\delta}_2 + \hat{\tau}_2 \phi(\mathbf{z}_i \hat{\delta}_2 / \hat{\tau}_2)$ and then use IVs $(\hat{m}_{i2}, \mathbf{z}_{i1})$ in the equation $y_{i1} = \alpha_1 y_{i2} + \mathbf{z}_{i1}\delta_1 + u_{i1}$ by IV.

- 5. (19.6) This is essentially given in equation (19.45), but were we allow the truncation points to depend on $\mathbf{x}_i$. Let y_i given $\mathbf{x}_i$ have density $f(y | \mathbf{x}_i, \beta, \gamma)$, where β is the vector indexing $E(y_i | \mathbf{x}_i)$ and γ is another set of parameters. Then the density of y_i given $\mathbf{x}_i$, $s_i = 1$, when $s_i = \mathbb{I}\{a_1(\mathbf{x}_i) < y_i < a_2(\mathbf{x}_i)\}$, is

$$p(y | \mathbf{x}_i, s_i = 1) = \frac{f(y | \mathbf{x}_i; \beta, \gamma)}{F(a_2(\mathbf{x}_i) | \mathbf{x}_i; \beta, \gamma) - F(a_1(\mathbf{x}_i) | \mathbf{x}_i; \beta, \gamma)}, \quad a_1(\mathbf{x}_i) < y_i < a_2(\mathbf{x}_i).$$

In the Hausman and Wise (1977) study, $y_i = \log(\text{income}_i)$, $a_1(\mathbf{x}_i) = -\infty$, and $a_2(\mathbf{x}_i)$ was a function of family size.